

Source: https://upload.wikimedia.org/wikipedia/commons/e/ee/Artificial_Neural_Network_with_Chip.jpg

24CS206 VIBE CODING (AI BASED APP DEVELOPMENT)

Hours Per Week :

L	T	P	SL	C
0	0	2	0	1

PREREQUISITE KNOWLEDGE: Basic computer operating skills.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces students to the fundamentals of AI-assisted application development. Students will learn how to use modern AI coding tools to design, develop, test, and improve simple applications by describing requirements in natural language. The course covers prompt writing, basic app design concepts, user interface creation, debugging with AI assistance, and responsible use of AI in software development. By the end of the course, students will be able to build simple applications while developing computational thinking, problem-solving, and collaboration skills.

MODULE-1

UNIT-1 **0L+0T+6P+0SL=6 Hours**

INTRODUCTION TO VIBE CODING AND AI-ASSISTED APP DEVELOPMENT

What is software? Types of applications, Mobile apps vs. Web apps vs, Desktop apps Real-world examples.

Low-code/No-code platforms, AI-assisted development, What is Vibe Coding? Pros & cons, Fundamentals of Vibe Coding.

Introduction to AI: What is Artificial Intelligence? ,What is Generative AI?, Large Language Models (LLMs), AI coding assistants

AI Tools for App Development: Introduce tools such as: ChatGPT ,GitHub Copilot, Cursor , Firebase Studio (or another beginner-friendly AI IDE)

Responsible AI: Ethics, Privacy, Academic Integrity, Copyright, and Safe Use of AI.

UNIT-2 **0L+0T+6P+0SL=6 Hours**

PROMPT ENGINEERING

Introduction to Prompt Engineering, What is a Prompt? Characteristics of an Effective Prompt. Types of Prompts: Instruction-based, Conversational, Role-based and Iterative Prompting.

Role of Prompt Engineering in AI-Assisted Development. Prompt Refinement Techniques for Improving AI Responses, Generate a sample image using an AI-generated prompt. Resume & Email Generation. AI Hallucinations in Code Generation and Mitigation Techniques.

MODULE-2

UNIT-1 **0L+0T+6P+0SL=6 Hours**

APPLICATION DEVELOPMENT APPROACHES

Introduction to Application Development. Traditional Software Development: Overview, characteristics, advantages and limitations. AI-Assisted Application Development: Fundamentals of Vibe Coding, role of Generative AI, AI coding assistants and AI development environments.

Introduction to AI Development Environments (AI IDEs). Exploring AI Development Tools: Replit, Cursor, Firebase Studio, GitHub Copilot and other beginner-friendly AI platforms.

Comparison of Traditional, Low-Code/No-Code and AI-Assisted Development. Future Trends in AI-Assisted Software Development and Career Opportunities.

UNIT-2 **0L+0T+6P+0SL=6 Hours**

FUNDAMENTALS OF APPLICATION DEVELOPMENT

Application Architecture: Components of an Application, Frontend (User Interface), Backend (Business Logic), Database, APIs and Client Server Architecture.

User Interface (UI) and User Experience (UX): Introduction to UI and UX, UI Components (Buttons, Text Fields, Images, Cards, Navigation Bars), Responsive Design Principles and AI-Assisted UI Generation.

PRACTICES:

Exercise 1: Creating Your First Web Page

Tasks:

Generate a basic HTML webpage using AI.

Add a title and heading.

Preview the webpage in a browser.

Analysis Task:

Identify the basic structure of an HTML document.

Exercise 2: Personal Profile Page

Tasks:

Generate a personal profile webpage.

Add name, photo, educational details and hobbies.

Modify the content using AI prompts.

Analysis Task:

Compare the generated page before and after customization.

Exercise 3: Creating Lists and Tables

Tasks:

Generate ordered and unordered lists.

Create a timetable using an HTML table.

Modify rows and columns using AI.

Analysis Task:

Identify suitable scenarios for using lists and tables.

Exercise 4: Working with Images and Hyperlinks

Tasks:

Add images to a webpage.

Create internal and external hyperlinks.

Generate a simple navigation menu.

Analysis Task:

Explain the role of images and hyperlinks in web pages.

UNIT-3 **0L+0T+6P+0SL=6 Hours**

FUNDAMENTALS OF APPLICATION DEVELOPMENT

Backend and Databases: Introduction to Backend, Roles of Backend, Introduction to Databases, SQL and NoSQL Databases, Cloud Databases (Overview), AI-Assisted Backend and Database Generation.

APIs and Cloud Services: Introduction to APIs, Request and Response Concept, Connecting Applications using APIs, Cloud Computing Basics, Backend-as-a-Service (BaaS), Cloud Platforms for AI-Assisted Development.

Application Deployment: Application Testing, Cloud Deployment, Publishing Web Applications, Introduction to Mobile Application Publishing, Version Updates and Maintenance.

PRACTICES:

Exercise 1: Personal Profile Web Page

Tasks:

Generate a personal profile webpage using AI.

Add headings, paragraphs, images and hyperlinks.

Modify the webpage content and layout.

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

Analysis Task:

Explain the HTML elements used in the webpage.

Exercise 2: Calculator Application

Tasks:

Generate a basic calculator using AI.

Perform addition, subtraction, multiplication and division.

Modify the interface and functionality using AI prompts.

Analysis Task:

Explain how AI generated the application and the modifications performed.

Exercise 3: To-Do Application

Tasks:

Generate a simple To-Do application using AI.

Add, edit and delete tasks.

Test the application with sample tasks.

Analysis Task:

Describe the workflow of the To-Do application.

COURSE OUTCOMES:

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2			2			3		2		2	3	
CO2	2	3	2		3			2	2	2		2	3	
CO3	2	2	3	2	3				2			2	3	3
CO4	2	3	2	3	3			2				2	3	3
CO5	2	2	3	2	3			3	3	3		3	3	3

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcome	Bloom's Level	Module No.	Mapping with POs
C1	Describe the concepts of AI, Generative AI, Vibe Coding, prompt engineering and AI-assisted software development.	Understand	Module 1	PO1, PO2
C2	Use AI coding assistants and prompt engineering techniques to create digital content and application prototypes.	Apply	Module 1 Unit 2, Module 2 Unit 1	PO2, PO3
C3	Build simple AI-assisted web applications using AI development environments and beginner-friendly development tools.	Apply	Module 2 Unit 1, 2 & 3	PO2, PO3, PO5
C4	Examine the architecture and workflow of modern applications including frontend, backend, databases, APIs and cloud deployment.	Analyze	Module 2 Unit 2 & 3	PO1, PO2, PO4
C5	Design, test and present simple AI-assisted applications by applying ethical AI practices and application development workflows.	Create	All Modules	PO2, PO3, PO4, PO5

MAPPING OF SDGS AND IKS

Unit	Topic	Sustainable Development Goals (SDGs)	Indian Knowledge System (IKS)
Unit 1	Introduction to Vibe Coding and AI-Assisted App Development	SDG 4 Quality EducationSDG 9 Industry, Innovation & Infrastructure	Encourages logical thinking, creativity and knowledge acquisition, reflecting the inquiry-based learning traditions of the Gurukul system and Nyaya philosophy.

Unit	Topic	Sustainable Development Goals (SDGs)	Indian Knowledge System (IKS)
Unit 2	Prompt Engineering	SDG 4 Quality EducationSDG 9 Industry, Innovation & Infrastructure	Effective communication with AI reflects the importance of structured questioning, dialogue and reasoning found in Nyaya (Indian School of Logic) and the Socratic-style discourse practiced in ancient Indian education.
Unit 3	Application Development Approaches and AI Development Tools	SDG 8 Decent Work & Economic GrowthSDG 9 Industry, Innovation & Infrastructure	Promotes innovation, skill development and technology adoption, aligning with India's long-standing tradition of craftsmanship, engineering and practical problem-solving.
Unit 4	Fundamentals of Application Development	SDG 9 Industry, Innovation & InfrastructureSDG 11 Sustainable Cities & Communities	Systematic organization of application components reflects the structured approach to knowledge organization found in ancient Indian sciences such as Vastu Shastra and Arthashastra.
Unit 5	Backend, APIs, Cloud Services and Application Deployment	SDG 9 Industry, Innovation & InfrastructureSDG 17 Partnerships for the Goals	Sharing applications and digital collaboration reflect the Indian philosophy of "Vasudhaiva Kutumbakam" (The World is One Family), emphasizing collaboration, knowledge sharing and collective progress through technology.

TEXT BOOKS:

Jon Duckett, HTML & CSS: Design and Build Websites, John Wiley & Sons, 1st Edition, 2011.

Julie C. Meloni, HTML, CSS, and JavaScript All in One, Sams Teach Yourself, 3rd Edition, Pearson Education, 2018.

David Flanagan, JavaScript: The Definitive Guide, 7th Edition, O'Reilly Media, 2020.

Leo Porter and Daniel Zingaro, Learn AI-Assisted Programming: A Beginner's Guide to Using ChatGPT, GitHub Copilot, and Generative AI, Manning Publications, 2024.

REFERENCE BOOKS:

Paul Deitel & Harvey Deitel, Internet & World Wide Web: How to Program, 5th Edition, Pearson.

Jon Duckett, Beginning Web Programming with HTML, XHTML, and CSS, Wiley.

David Flanagan, JavaScript: The Definitive Guide, O'Reilly.

Chris Minnick, Responsive Web Design with HTML5 & CSS, Wiley.

Packt Publishing, Generative AI for Beginners, Latest Edition